

The Config Bench

V8 — Qucs Schematic Build Guide

Six circuits | Component values | Netlists | Export instructions | @TheConfigBench

How to use this guide: Each circuit has a step-by-step Qucs build sequence, component values, and the DC operating point simulation setup. Build each schematic, run the simulation to verify your answer key values, then export as PNG (File → Export → PNG, 150dpi minimum) for import into Xournal++. **Q1-Q3:** Simple enough for Xournal++ hand-draw if preferred. **Q4-Q6:** Use Qucs — parallel nodes need to be drawn cleanly.

Export Settings for Xournal++ Import

File → Export Document as Image | Format: PNG | Resolution: 150 dpi minimum (200 preferred) | Background: White or transparent | In Xournal++: Insert → Image → select PNG → landscape page, dark background. Resize to fill left 60% of page — leave right 40% for handwritten math.

Q1 — Single Resistor | 120Ω / 6V

V=6V ■ R=120 Ω ■ Find: I ■ Answer: 50mA

Qucs Build Steps:

1. New schematic (Ctrl+N)
2. Place DC Voltage Source (V1): Value = 6V. Positive terminal up.
3. Place Resistor (R1): Value = 120 Ohm. Wire in series with V1.
4. Place Ground (GND) on negative terminal of V1.
5. Add wire connecting R1 top to V1 positive, R1 bottom to GND.
6. Add DC operating point simulation component (Simulate → DC simulation).
7. Simulate (F2). Node voltage at R1 top = 6V. Confirm I through R1 = 50mA.

Netlist reference:

```
# Wire: V1(+) → R1(top) → node_A
# Wire: R1(bottom) → GND
# Wire: V1(-) → GND
<.DC DC1 "yes" "" "" "">
```

Q2 — Series Circuit | 100Ω + 150Ω / 12V

V=12V ■ R1=100 Ω R2=150 Ω in series ■ Find: I, V_R1, V_R2 ■ Answers: 48mA / 4.8V / 7.2V

Qucs Build Steps:

1. New schematic.
2. Place DC Voltage Source (V1): Value = 12V.
3. Place R1 = 100 Ohm in series above V1 positive terminal.
4. Place R2 = 150 Ohm in series above R1 (top of R1 connects to bottom of R2).
5. Wire top of R2 back to V1 positive (completing the loop) — or use a node label.
6. Place GND on V1 negative.
7. Add DC simulation. Run (F2).
8. Check: node between R1/R2 shows voltage divider point (~7.2V from GND). Current = 48mA.

Netlist reference:

```
# Wire: V1(+) → R1(top) → node_A
# Wire: node_A → R2(top) ... wait: series means R1 bottom → R2 top → node_B
# Corrected: V1(+) → R1(top), R1(bottom) → node_mid, node_mid → R2(top), R2(bottom) → GND,
V1(-) → GND
<.DC DC1 "yes" "" "" "">
```

Q3 — Series: Find Missing R | 9V / 30mA / R1=100Ω

V=9V ■ I=30mA ■ R1=100Ω ■ Find: R2 ■ Answer: R2 = 200Ω

Qucs Build Steps:

1. New schematic.
2. Place DC Voltage Source V1 = 9V.
3. Place R1 = 100 Ohm in series.
4. Place R2 = 200 Ohm in series (this is the answer students must find).
5. GND on V1 negative.
6. Run DC simulation. Confirm I = 30mA through the loop.
7. On-camera: Build with R2 as unknown (?), solve on Xournal++, then reveal in Qucs.

Netlist reference:

```
# This is the answer – students solve for this
# Series: V1(+) → R1 → R2 → GND → V1(-)
<.DC DC1 "yes" "" "" "">
```

Q4 — Parallel Circuit | 220Ω ■ 330Ω / 9V

$V=9V$ ■ $R1=220\Omega$ ■ $R2=330\Omega$ ■ Find: R_{total} , I_{total} , I_{R1} , I_{R2} ■ Answers: 132Ω / $68.2mA$ / $40.9mA$ / $27.3mA$

Qucs Build Steps:

1. New schematic.
2. Place DC Voltage Source $V1 = 9V$. Positive terminal connects to top node (node_TOP).
3. Place $R1 = 220\text{ Ohm}$: top connects to node_TOP, bottom connects to node_BOT (GND).
4. Place $R2 = 330\text{ Ohm}$: top connects to node_TOP, bottom connects to node_BOT (GND).
5. Place GND at node_BOT. Connect $V1$ negative to node_BOT.
6. Both resistors share the same top and bottom nodes — that is the parallel connection.
7. Run DC simulation. Check: node_TOP = 9V. I through $R1 = 40.9mA$, I through $R2 = 27.3mA$.
8. Total current from $V1 = 68.2mA$.

Netlist reference:

```
# Parallel: both R1 and R2 connect between node_TOP and GND
# V1(+) → node_TOP, V1(-) → GND
# R1: node_TOP → GND
# R2: node_TOP → GND
<.DC DC1 "yes" "" "" "">
```

Q5 — Parallel: Find Supply Voltage | 560Ω ■ 560Ω / $I=64.3mA$

$I_{total}=64.3mA$ ■ $R1=R2=560\Omega$ in parallel ■ Find: V ■ Answer: 18V

Qucs Build Steps:

1. New schematic.
2. Place DC Voltage Source $V1 = 18V$ (this is the answer — students solve for it).
3. Place $R1 = 560\text{ Ohm}$ and $R2 = 560\text{ Ohm}$ in parallel (both between node_TOP and GND).
4. Run DC simulation. Confirm $I_{total} = 64.3mA$.
5. On-camera approach: Present the circuit with $V1$ as unknown. Solve $V = I \times R_{total}$ on Xournal++, then plug 18V into Qucs to verify.

Netlist reference:

```
# Answer value – solve for this on Xournal++ first

# Parallel: R1 and R2 both between node_TOP and GND
# V1(+) → node_TOP, V1(-) → GND
```

```
<.DC DC1 "yes" "" "" "">
```

Q6 — CURVEBALL: Series-Parallel | $68\Omega + (100\Omega \parallel 100\Omega) / 12V$

$V=12V$ ■ $R1=68\Omega$ (series) ■ $R2=100\Omega$ ■ $R3=100\Omega$ (parallel pair) ■ Answers: $R_{total}=118\Omega$ / $I_{total}=101.7mA$ / $V_{R1}=6.91V$ / $V_{parallel}=5.09V$ / $I_{branch}=50.9mA$ each

Qucs Build Steps:

1. New schematic.
2. Place DC Voltage Source V1 = 12V. Positive terminal to node_A.
3. Place R1 = 68 Ohm: top at node_A, bottom at node_B. This is the series resistor.
4. node_B is the junction point — this is where current enters the parallel pair.
5. Place R2 = 100 Ohm: top at node_B, bottom at node_C (GND).
6. Place R3 = 100 Ohm: top at node_B, bottom at node_C (GND).
7. Place GND at node_C. Connect V1 negative to node_C.
8. Run DC simulation.
9. Check node_A = 12V, node_B = 5.09V, node_C = 0V (GND).
10. V across R1 = $12 - 5.09 = 6.91V$. Current through R1 = 101.7mA. Each branch = 50.9mA.

Netlist reference:

```
# Series resistor – connects node_A to node_B
# Parallel branch 1 – connects node_B to GND
# Parallel branch 2 – connects node_B to GND

# node_A = V1(+) = 12V
# node_B = junction of R1(bottom), R2(top), R3(top) = 5.09V
# node_C = GND = V1(-) = R2(bottom) = R3(bottom) = 0V

<.DC DC1 "yes" "" "" "">
```

Key teaching moment for Q6: Point out node_B on the schematic. That is where the series world ends and the parallel world begins. The voltage at node_B (5.09V) is what drives both parallel branches — not the full 12V. R1 consumed 6.91V just getting the current to the junction. This is real circuit analysis — every practical circuit has both series and parallel elements.